

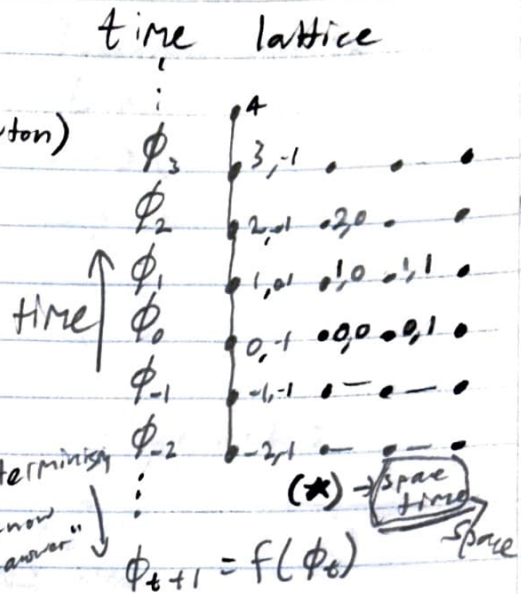
Chaotic Field Theory

The Law: $F_z[\phi^c] = 0$

Conventional: time is separate (Newton)

→ we accept that there is an equivariance of our law: it stays the same for all time

→ Hydraulics: theory of how fluids behave for the working Man (technicians were "steam bros")



$$\Rightarrow F_t[\Phi] = 0, \quad f(\phi_t) = 0$$

→ introduce new dimension: a lattice in space-time (we introduce one, but including 3 would be Yang-Mills theory)

→ Our law is described at every location on the lattice by a kinetic (adjust to neighbors) + a potential (force dependent on location)

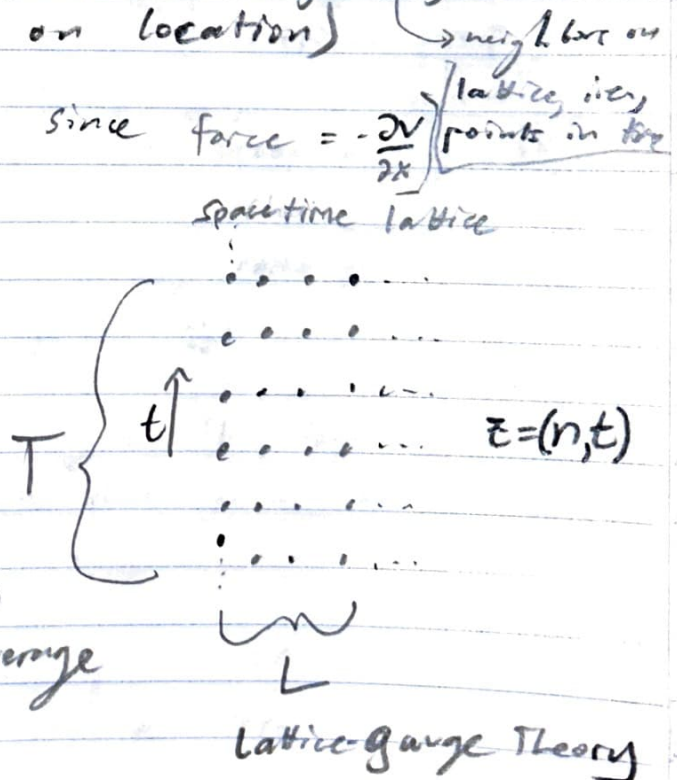
kinetic (adjust to neighbors)

$$F_z = K + V[\phi_z]$$

→ parameter $r = \frac{\|K\|}{\|V\|}$

→ $T \times L$ represents a finite volume in the spacetime double-lattice

→ we use these volumes in practice to approximately average over the law:



→ Monte-Carlo Methods

$$\langle a \rangle_{LT} \approx \sum_{t=1}^T \sum_{z=1}^L p_z a_z$$

Thermodynamic Limit

$$T \rightarrow \infty$$

$$L \rightarrow \infty$$

→ p_z estimates the "importance" of a particular configuration wrt others

→ This is all the 'conventional' approach:

Conventional

$$\langle a \rangle_{L,T} \approx \sum_{t=1}^T \sum_{z=1}^L p_z a_z$$



Symmetry

Thermodynamic Limit $T, L \rightarrow \infty$

→ Cyclic group all subgroups of H
 $H: C_3$

$$\langle a \rangle = \sum_{G/H} \langle a \rangle_{G/H}$$

← quotient group

→ we know by looking at, e.g., the clouds, that the natural order does not necessarily fall into a symmetric state:

→ along with a symmetry, comes a cascading tree of broken symmetry

→ we start with the symmetry of the law: the simplest case is that it applies always & anywhere, i.e., translationally invariant

→ Consider a Period-3 equilibrium solution: this is represented by a translation of the lattice

→ if we have a symmetry, then we can break our state space into a fundamental unit of the symmetry and use this to construct solutions

↓ subgroups of the symmetry

Cyclic group: $C_3, C_4, \dots, C_{27}$

→ in this way, you already start in the thermodynamic limit

→ For time, this was developed by

• Gutzwiller (QM) } developed a theory of

• Ruelle (stat mech)

• Cvitanovic: "I spent good time w/ both of them and they both think their theory has nothing to do with each others"

→ Sometimes periodicities arise in space:

space groups: translationally invariant solutions

point groups: rotations, inversions, reflections

time groups: periodic solutions

→ everything you compute is exact: already in the infinite-time limit using symmetry groups

→ In Practice: Symmetry Group Method

• We need a sense of "neighborhood" size.

Consider the shift operator: essentially a re-indexing

Operators
shift:

$$(\Pi \phi)_z = \Phi_{z+r}$$

↑
coordinate

relabeling each point on the spacetime lattice

derivative:

$$(\partial_i \phi)_z = \frac{\phi_{z+1} - \phi_z}{\Delta t}$$

since we are on a discrete lattice, $\Delta t \equiv 1$

$$\Rightarrow (\partial_i \phi)_z = (\Pi - \mathbb{I}) \phi_z$$

identity operator

⇒ Our derivative operator changes from local to global (info about neighbors)

$$F[\Phi]_z = \underbrace{\sum_i (\Pi - \mathbb{I}) \cdots \phi_z}_{\text{kinetic}} + \underbrace{V[\Phi]}_{\text{potential}}$$

Deterministic state space (M, F)

↳ we consider only the states which obey the law, in general

↳ (M_{det}, F)

$$M_{\text{det}} = \{ \phi^{\text{classical}} \}$$

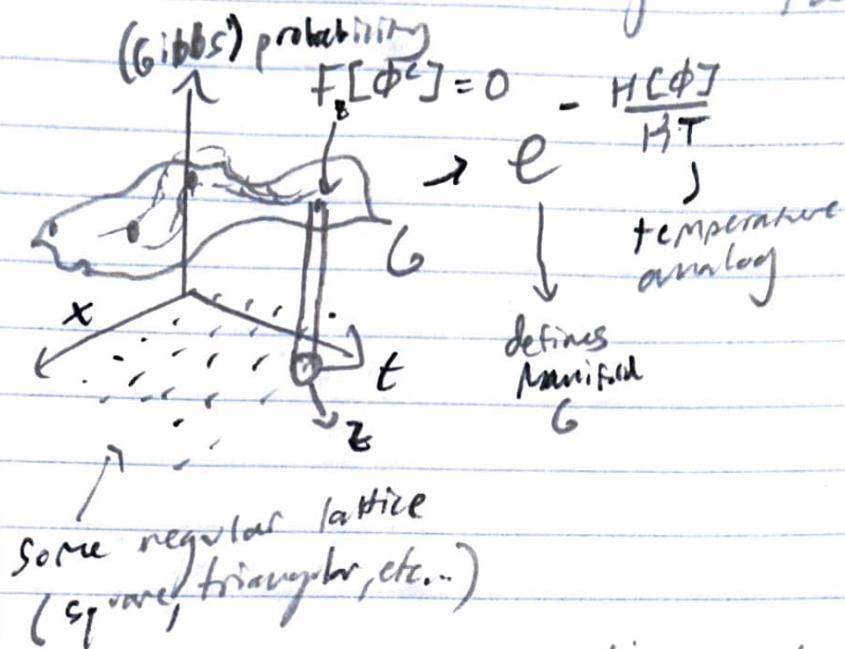
↳ infinitely many

→ Now comes the 'thinking part':

use symmetry to organize the infinitely many solutions

→ in chaotic / unstable systems, we lose information exponentially fast : hence, we stick to stable, exact solutions

→ but how do we weight the solutions?



→ the most important points are the extrema of the manifold G . Hence, the tangent bundle / derivative of G is the relevant quantity, ∂G

high dimensional functional

$$P[\phi] = \mathcal{S}[F[\phi]] \rightarrow \log \text{ of our configuration}$$

$$P = \int_{M_c} d\phi \delta(F[\phi]) \rightarrow \text{Jacobians of derivatives of functions (integral of } \vec{G})$$

tells us the importance of different solutions

→ we start with small period solutions and increase from there

$$P = \frac{1}{\|\det J\|}, \quad J = \frac{\partial F}{\partial \phi}$$

→ In stat mech, when we have lots of solutions, we write the partition function

→ For infinitely local limit, the probability of any configuration at a pair is a product of the individual probabilities

→ But we are better at addition than multiplication,

$$Z = \sum_{\phi \in M_{\text{int}}} e^{S[\phi]}$$

→ Strong coupling involves including the influence of neighbors

→ Chaos is a strongly-coupled, Euclidean Field Theory

→ Chaos is on the opposite end of state space types* as harmonic oscillators

→ Kivelson Seminar Putgers Math